



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAR PARKING TOKEN SYSTEM

Car parking Token System Using Queue (ADT)

A car parking token system organizes customers so everyone is served in the order they arrive — avoiding confusion, reducing waiting-time disputes, and improving service efficiency.

The Queue Abstract Data Type (ADT) is the most suitable choice because it follows the **First In, First Out (FIFO)** principle — the customer who arrives first is served first.

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What is a Queue?

A **Queue** is a linear data structure that follows the FIFO (First In, First Out) principle.

1

Enqueue

Add a customer to the rear of the queue.

2

Dequeue

Remove the customer from the front of the queue.

3

Front (Peek)

View the first customer.

4

IsEmpty

Check whether the queue is empty.

EXAMPLE



Service Order

101 → 102 → 103 → 104

JUSTIFICATION

Why Queue is Suitable

- ✓ Ensures customers are served in arrival order
- ✓ Prevents customers from skipping the line
- ✓ Simple and easy to implement
- ✓ Provides fair and organized service
- ✓ Suitable for cafeterias, banks, hospitals, and ticket counters

TIME COMPLEXITY

Operation	Complexity
Enqueue	$O(1)$
Dequeue	$O(1)$
Front	$O(1)$

Conclusion: The Queue ADT guarantees fairness and efficient customer service using the FIFO principle.

BEYOND THE CAFETERIA

Real-World Applications



Cafeteria token systems



Bank counters



Hospital registration



Railway ticket booking



Printer scheduling



Call centers

Thank You

